SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA51
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: **Total Marks: 50**

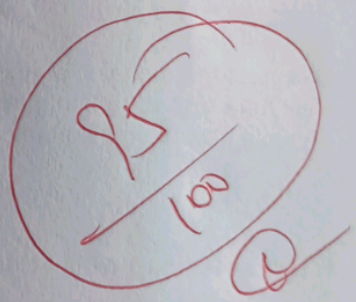
Annexure A

Code of Conduct:

I G.Ram srinivas. *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

G.Ram



Annexure A
Analytical Problem Solving

1. RSA Key Generation and Security Analysis
Given:
$p = 17, q = 19$

Compute n and $\phi(n)$
Choose $e = 5$ and calculate the private key d
Encrypt the message $M = 7$
Decrypt the ciphertext and verify correctness
Analyze the impact of choosing a small prime number on RSA security

2. ElGamal Encryption Analysis
Given:
$p = 23$, $g = 5$, private key $x = 7$, random $k = 3$, message $M = 10$

Compute the public key
Perform encryption to generate ciphertext pair (C1, C2)
Decrypt the ciphertext to retrieve the original message
Analyze how randomness (k) improves security in ElGamal

3. RSA Digital Signature Verification
Given:
$p = 11$, $q = 17$, $e = 7$, message $M = 8$

Compute n and $\phi(n)$
Find private key d
Generate the digital signature for the message
Verify the signature using the public key
Analyze how digital signatures ensure non-repudiation

4. Diffie-Hellman with Attack Resistance Evaluation
Given:
$p = 29$, $g = 2$
User A private key = 5
User B private key = 12

Compute public keys for both users
Compute the shared secret key
Analyze what happens if an attacker intercepts and replaces public keys
Suggest a cryptographic solution to ensure authenticity

5. Hybrid Encryption System Analysis
Given:
A system uses RSA for key exchange and AES for data encryption
RSA encryption time = 5 ms per key exchange
AES encryption time = 0.5 ms per data block
Total data blocks = 500

Calculate total time taken for encryption
Compare with using only RSA for encrypting all data
Analyze why hybrid encryption is more efficient
Justify its use in modern secure communication systems

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate	Selects correct	Inappropriate or partially	Unable to select

		method/formula with strong justification	method with minor errors	correct method	method
Solution Process	25% 20	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20% 20	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15% 15	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

85

CO2-AT1

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CSA5101 - Cryptography and network security.

1A) RSA key Generation and security Analysis

given:-

$p = 17, q = 19$

public exponente = 5

Message $M = 7$

RSA modulus:-

$n = p \times q$

$= 17 \times 9$

$\boxed{n = 323}$

Euler's totient function:-

$\phi(n) = (p-1)(q-1)$

$= 16 \times 18$

$\boxed{\phi(n) = 288}$

private ked d

$ed \equiv 1 \pmod{\phi(n)}$

$5d \equiv 1 \pmod{288}$

$288 = 5(57) + 3$

$5 = 3(1) + 2$

$3 = 2(1) + 1$

work backward.

$1 = 2(3) - 5$

$1 = 3(288) - 5(57) - 5$

$1 = 3(288) - 5(172)$

$d = -172 \pmod{288}$

$\boxed{d = 116}$

Encrypt the message:-

$$c = M^e \bmod n$$

$$c = 7^5 \bmod 323$$

$$7^2 = 49, \quad 7^4 = 2401 \equiv 140 \pmod{323}$$

$$7^5 \equiv 140 \times 7 = 980 \equiv 11 \pmod{323}$$

the ciphertext $C = 11$

for verification $M = c^d \bmod n$

$$m = 11^{173} \bmod 323$$

$$\boxed{M = 7}$$

using small primes such as 17 and 19 makes the RSA modulus $n = 323$ extremely easy to factor

A- ElGamal Encryption Analysis :-

given:-

prime number $p = 23$

$g = 5$

$x = 7$

$k = 3$

$M = 10$

calculate public key:-

$$y = g^x \bmod p$$

$$= 5^7 \bmod 23$$

$$y = 4 \times 2 \times 5 = 40$$

$\boxed{y = 17}$

∴ public key $(p, g, y) = (23, 5, 17)$

private key: $x = 7$

Encrypt the message:-

$$c_1 = g^k \bmod p$$

$$c_1 = 5^3 \bmod 23$$

$$= 125 \bmod 23$$

$\boxed{c_1 = 10}$

$$s = y^k \bmod p$$

$$s = 17^3 \bmod 23$$

$$17^2 \bmod 23 = 13$$

$$s = 13 \times 17 \bmod 23 = 221 \bmod 23$$

$\boxed{s = 14}$

$$c_2 = M \times s \bmod p$$

$$c_2 = 10 \times 14 \bmod 23$$

$$= 140 \bmod 23$$

$\boxed{c_2 = 2}$

the ciphertext pair $(c_1, c_2) = (10, 2)$

Decrypt the ciphertext:-

$s \equiv c_1^x \bmod p$

$s = 10^7 \bmod 23 = 14$

$14 \times 5 \bmod 23 = 1$

$M = c_2 \times s^{-1} \bmod p$

$M = 2 \times 5 \bmod 23$

$\boxed{M = 10}$

A fresh unpredictable k should be used for every encryption. Reusing k can reveal relationship b/w message.

ElGamal improves confidentiality through randomized encryption

4) RSA Algorithm signature verification:-

$p = 11$

$q = 17$

public exponent $e = 7$

message $M = 8$

$n = pq$

$= 11 \times 17$

$\boxed{n = 187}$

$\phi(n) = (p-1)(q-1)$

$= 10 \times 16$

$\boxed{\phi(n) = 160}$

ffie-Hellman key Exchange with Attack Resistance Evalution:

$p = 29$

$g = 2$

$a = 5$

$b = 12$

public key $A = g^a \bmod p$

$= 2^5 \bmod 29 = 32 \bmod 29$

$\boxed{A = 3.}$

$B = g^b \bmod p$

$= 2^{12} \bmod 29$

$2^{12} = 4096$

$4096 \bmod 29 = 7$

$\boxed{B = 7}$

alculate shared secret key:-

$K_A = B^a \bmod p$

$K_A = 7^5 \bmod 29$

$7^2 \bmod 29 = 20$

$7^4 \bmod 29 = 23$

$K_A = 23 \times 7 \bmod 29$

$\boxed{K_A = 16}$

$K_B = A^b \bmod p$

$K_B = 3^{12} \bmod 29$

$3^4 \bmod 29 = 23$

$3^8 \bmod 29 = 7$

Find the private key d

$ed \equiv 1 \pmod{\phi(n)}$

$7d \equiv 1 \pmod{\phi(n)}$

$161 \bmod 160 = 1$

$\boxed{d = 23}$

Generate the digital signature:-

$S = M^d \bmod n$

$S = 8^{23} \bmod 187$

$8^2 \bmod 187 = 64$

$8^4 \bmod 187 = 169$

$8^8 \bmod 187 = 137$

$8^{16} \bmod 187 = 69$

since $23 = 16+4+2+1$

$S = 8^{16} \times 8^4 \times 8^2 \times 8 \bmod 187$

$\boxed{S = 83}$

verify signature:-

$M' = S^e \bmod n$

$= 83^7 \bmod 187$

$\boxed{M' = 8}$

gital signature provide the security properties like thentication, integrity, Non-repudiation

$K_B = 3^8 \times 3^4 \bmod 29$

$K_B = 7 \times 23 \bmod 29$

$\boxed{K_B = 16}$

$\boxed{K_A = K_B = 16}$

Both users independently obtain the same shared secret.

- Analyze a man-in-the-middle attack

By itself does not authenticate the exchanged public keys. An attacker can intercept the public keys and replace them with their own values.

Diffie-Hellman establishes a shared secret, but authentication is necessary to prevent an attacker from substituting public keys.

Hybrid Encryption system Analysis :-

- RSA key exchange time = 5 ms / key
- AES encryption time = 0.5 ms
- Total data blocks = 500

Total AES encryption time is

$$T_{AES} = 500 \times 0.5$$

$$\boxed{T_{AES} = 250 \text{ ms}}$$

Total hybrid encryption time

$$T_{hybrid} = T_{RSA} + T_{AES}$$

$$= 5 + 250 \quad = 255 \text{ ms}$$

compare with RSA-only encryption.

$$T_{RSAonly} = 500 \times 5$$

$$\boxed{T_{RSA\text{-}only} = 2500\ ms}$$

comparison:

parameter	Hybrid encryption	RSA-only
RSA operations	1 key exchange	500 blocks encryption
AES data blocks	500	0
Total time	255ms	2500 ms
main data encryption	AES	RSA

Under this assumption time

$$2500 - 255 = 2245\ ms$$

percentage reduction $\frac{2500-255}{2500} \times 100 = 89.8\%$

RSA securely transfer or protects the AES session key

AES encrypts large amount of data efficiently

AES support efficient bulk encryption. whereas RSA is computationally expensive and is not designed to encrypted arbitrary large data directly.